

DATA-620 FORMAL SEMANTIC EXPERIMENT PLAN

Experimental parameter		Planned response
Experimenter(s)		Roxy Alexander
Research question		What model (medical, moral, social) of disability shows up the most in the documents linked within Stanford Encyclopedia of Philosophy (SEP)?
Motivation		To see if SEP reflects the current view of disability in Western Society, which is the medical model.
Variables	AI systems to be tested	Stanford Encyclopedia of Philosophy
	Controls held constant	- Search Platform (duckduckgo on Firefox) - Researcher (Roxy) - Computer (Roxy's Personal Mac) - Year (2026)
	Controls changed (explanatory variables)	Search words (key words) - Affliction - Ailment - Condition - Defect - Disablement - Disability - Disabled - Disease - Disorder - Handicap - Illness - Impairment - Inability - Incapacity - Incapability - Incapable - Infirmity - Invalidity - Malady - Unfitness - Weakness

	Response variable(s)	The Model of Disability (moral, medical, social): The Medical Model= The medical model of disability views disabilities as defects or abnormalities within an individual that need to be cured or managed through medical intervention. It focuses on the individual's physical condition rather than the societal barriers that may limit their participation in everyday life. The Social Model= The social model of disability suggests that people are disabled not by their impairments but by societal barriers, negative attitudes, and lack of accessibility. It emphasizes the need to remove these barriers to create equality and independence for disabled individuals. The Moral Model= The moral model of disability views disability as a result of moral failings or punishment for wrongdoing, often linked to religious beliefs. This perspective can lead to stigma and shame for individuals with disabilities, as it places the responsibility for their condition on their character or actions.
	Potential confounders	As a disability scholar conducting this specific experiment on the Stanford Encyclopedia of Philosophy (SEP), identifying confounders is critical to ensuring your results are seen as rigorous rather than subjective. A confounder is an outside influence that changes the effect of a dependent and independent variable. In your study, these variables might "skew" why a certain article is labeled Medical, Social, or Moral, regardless of the keyword used. 1. Historical Context (The "Temporal" Confounder) The SEP contains entries written over several decades. The Issue: An article written in 1995 is much more likely to use the Medical Model or Moral Model by default, simply because the Social Model had not

		yet gained mainstream academic traction in philosophy. The Impact: You might find a high Medical Model count not because the <i>keyword</i> is inherently "medical," but because the <i>article</i> is old. 2. Disciplinary Sub-field (The "Domain" Confounder) The SEP covers everything from Logic to Bioethics to Ancient Religion. The Issue: If a keyword like "Weakness" appears in an article about <i>Ancient Greek Ethics</i>, it will almost certainly be coded as the Moral Model (). If it appears in an article about <i>Geriatrics</i>, it will be coded as Medical. The Impact: The "model" might be a result of the philosophical sub-discipline (Ethics vs. Science) rather than the keyword itself. 3. Polysemy (The "Dual-Meaning" Confounder) Many of your keywords have non-disability meanings in philosophy. The Issue: "Invalidity" is a core term in Formal Logic (an invalid argument). "Condition" is a core term in Metaphysics (necessary and sufficient conditions). The Impact: If these "logical" uses are included in your tally, they could create "noise" in your data. (In our experiment, we filtered these out, but in a raw automated search, this is a major confounder). 4. Authorial Bias (The "Standpoint" Confounder) The SEP entries are written by individual experts who may have a personal or professional bias. The Issue: A scholar trained in Critical Disability Theory will frame "Impairment" through a Social Model lens, whereas a Bioethicist might frame the exact same word through a Medical Model lens.
--	--	---

		The Impact: The result tells us more about the <i>author's background</i> than the <i>keyword's inherent meaning</i>. 5. The "One-Model-Per-Article" Restriction (The "Reductionist" Confounder) Your own experimental rule is a potential source of bias. The Issue: Most modern philosophical articles (like <i>Disability: Definitions and Models</i>) actually discuss all three models to provide a balanced overview. The Impact: By forcing a "strict one-model" rule, you are making a subjective judgment call on which model is "dominant." Another researcher might read the same article and choose a different dominant model.
Experimental method		Step 1: Keyword Selection Identify a comprehensive list of terms used in philosophy to describe human variation, limitation, and health. For this experiment, 21 keywords were selected, ranging from modern political identifiers (<i>Disability</i>) to historical moral descriptors (<i>Infirmity</i>) and clinical labels (<i>Disorder</i>). Step 2: Database Querying Access the Stanford Encyclopedia of Philosophy (SEP) search engine. For each keyword: Enter the term into the search bar. Systematically collect the results across multiple search pages (up to 10 pages or 100 articles per keyword) to ensure a representative sample of the philosophical corpus. Step 3: Systematic Literature Review Read each article or the specific section where the keyword appears. The goal is to identify the ontological framing of the condition described.

Step 4: Categorical Assignment (The "Rule of One")

Apply a strict one-model-per-article rule. Even if an entry mentions multiple viewpoints, it must be assigned to the single most dominant framework used by the author:

The Medical Model: If the article treats the condition as an internal "defect" or "abnormality" requiring a clinical cure or management.

The Social Model: If the article treats the condition as a neutral trait that becomes a disability only due to "societal barriers" and "lack of accessibility."

The Moral Model: If the article treats the condition as a reflection of "character," "moral failing," or "spiritual trial."

Step 5: Data Recording and Tallying

Record the results in a frequency table. Each keyword is assigned a raw count for each of the three models. This allows for a comparison of which terms "pull" the database toward specific ideological frameworks.

Step 6: Statistical Validation

Conduct a Pearson's Chi-squared (χ^2) Test of Independence.

Purpose: To determine if the distribution of models across keywords is due to chance or if there is a statistically significant correlation between the choice of vocabulary and the philosophical model employed.

Significance Threshold: A p-value of .05 is required to confirm that terminology dictates the framing of disability in the SEP.

Step 7: Result Interpretation

Analyze the data to identify "Terminological Bias." For example, if "Disease" triggers the Medical Model 88% of the time while "Disability" triggers the Social Model 85% of the time, the experiment

	concludes that the language chosen by a philosopher pre-determines the ethical and political status of the disabled subject.
Experimental null hypothesis	Differing search words should not affect the percentages of models represented.
Anticipated hypothesis test	Chi ² Test
Minimum sample size	21 key words and 3 models = ≥63
Interpretation of significant finding	The extremely small p-value (3.13×10^{-200}) indicates that the distribution of disability models across the 21 keywords is not random. This means the wording used to describe a condition in the Stanford Encyclopedia of Philosophy strongly predicts which philosophical model (Medical, Social, or Moral) will be applied in the corresponding article. In other words, terminology itself guides the conceptual framing. Keywords with clinical or pathological connotations (e.g., Disease, Disorder) overwhelmingly align with the Medical Model, while politically charged or socially grounded terms (e.g., Disability, Disablement) align with the Social Model. Words historically tied to virtue, agency, or personal character (e.g., Weakness, Inability) show high association with the Moral Model. The significant finding therefore suggests that the SEP does not treat disability-related concepts neutrally; instead, linguistic choice embeds and reproduces philosophical bias, shaping the moral, political, and medical narratives surrounding disability.
Limitations (especially consider confounders; offer potential mitigation strategies though they need not be used)	1. Search Engine Bias Different search engines index pages differently; even SEP's built-in search tool may not display results uniformly. Mitigation: Use multiple search engines or SEP's API (if available) to cross-check keyword results. 2. Researcher Subjectivity in Model Assignment

	Determining the “dominant model” in an article requires judgment, which introduces interpretive bias. Mitigation: Use a second coder and calculate inter-rater reliability (e.g., Cohen’s kappa) to validate consistency. 3. Keyword Overlap and Ambiguity Some keywords may be used metaphorically, historically, or outside the context of disability, potentially misclassifying articles. Mitigation: Establish strict inclusion/exclusion criteria or code ambiguous cases separately. 4. SEP’s Philosophical Scope SEP reflects the concerns and styles of analytic philosophy, not necessarily broader social or interdisciplinary perspectives on disability. Mitigation: Supplement SEP with additional philosophy or ethics databases to check for consistency. 5. Temporal Bias SEP entries are updated at different times—older articles may use outdated terminology or theoretical frameworks. Mitigation: Record each article’s revision date and assess whether model distribution changes over time. 6. Article Depth Variation Some entries might mention a keyword only briefly, while others explore it extensively, which can skew categorization. Mitigation: Add a weighting system for depth of discussion (e.g., number of mentions or section-level relevance). 7. Confounding by Topic Some keywords may appear primarily in specific philosophical subfields (bioethics vs. Theology vs. Metaphysics), which naturally favor different models.
--	--

	Mitigation: Categorize articles by subfield and perform stratified Chi-square tests.
Data Analysis	Chi-squared Test Results X² Statistic: 1079.20 Degrees of Freedom (df): 40 P-value: 3.13 X 10⁻²⁰⁰ (Approaching 0)
Interpretation	The results of the test are highly statistically significant. Rejection of the Null Hypothesis: Since the p-value is far below the standard threshold of (and even), we reject the null hypothesis that the keywords and the disability models are independent. Significant Association: This confirms that the choice of terminology in philosophical discourse is not neutral. Certain keywords (like "Disease" or "Disorder") are strongly associated with the Medical Model, while others (like "Disability" or "Disablement") are almost exclusively linked to the Social Model. Historical or agency-related terms (like "Weakness" or "Incapable") are significantly correlated with the Moral Model. Effect Size: The very large Chi-squared statistic relative to the degrees of freedom indicates a strong divergence between the observed counts and what would be expected if the keywords were distributed randomly across the models. In summary, the vocabulary used in the Stanford Encyclopedia of Philosophy acts as a reliable predictor of the philosophical framework (Medical, Social, or Moral) being employed to discuss human conditions.

DATA

Key Word	Medical Model (Pathologization of disability)	Moral Model (Reflection of one's moral actions)	Social Model (A result of the built world and society)	Total # of Articles
Affliction	12	14	0	26
Ailment	14	3	1	18
Condition	78	3	19	100
Defect	84	12	4	100
Disablement	1	0	9	10
Disability	12	3	85	100
Disabled	18	3	79	100
Disease	88	4	8	100
Disorder	83	3	14	100
Handicap	4	1	13	18
Illness	76	9	15	100
Impairment	38	0	17	55
Inability	71	25	4	100
Incapacity	39	9	2	50
Incapability	3	6	1	10
Incapable	42	55	3	100
Infirmity	4	6	0	10
Invalidity	34	0	3	37
Malady	12	2	1	15
Unfitness	7	3	0	10
Weakness	14	86	0	100
Sum	734	247	278	1259